3

Exercício 3.1

```
f[x_, y_] = x^2 y;
```

a) $\partial_x f(0,0) = 0$
b) $\partial_x f(x_0, y_0) = 2 x_0 y_0$
c) $\partial_y f(1,2) = 1$
d) $\partial_y f(x_0, y_0) = x_0^2$

Exercício 3.2

a)

```
f[x_, y_] = 3 x^2 + 2 y^2;
```

$\partial_x f = 6 x$
$\partial_y f = 4 y$

b)

```
f[x_, y_] = Sin[x^2 - 3 x y];
```

$\partial_x f = (2 x - 3 y) \text{Cos}[x^2 - 3 x y]$
$\partial_y f = -3 x \text{Cos}[x^2 - 3 x y]$

c)

```
f[x_, y_] = x^2 y^2 Exp[2 x y];
```

$\partial_x f = 2 e^{2 x y} x y^2 + 2 e^{2 x y} x^2 y^3$
$\partial_y f = 2 e^{2 x y} x^2 y + 2 e^{2 x y} x^3 y^2$

d)

```
f[x_, y_] = Exp[x] Log[x y];
```

$\partial_x f = \frac{e^x}{x} + e^x \text{Log}[x y]$
$\partial_y f = \frac{e^x}{y}$

e)

```
f[x_, y_] = Exp[Sin[x Sqrt[y]]];
```

$\partial_x f = e^{\text{Sin}[x \sqrt{y}]} \sqrt{y} \text{ Cos}[x \sqrt{y}]$
$\partial_y f = \frac{e^{\text{Sin}[x \sqrt{y}]} x \text{ Cos}[x \sqrt{y}]}{2 \sqrt{y}}$

f)

```
f[x_, y_] = (x^2 + y^2)/(x^2 - y^2);
```

$\partial_x f = -\frac{4 x y^2}{(x^2 - y^2)^2}$
$\partial_y f = \frac{4 x^2 y}{(x^2 - y^2)^2}$

g)

```
f[x_, y_] = x Cos[x] Cos[y];
```

$\partial_x f =$ Cos[y] (Cos[x]-x Sin[x])
$\partial_y f =$ -x Cos[x] Sin[y]

h)

```
f[x_, y_] = ArcTan[x^2 y^3];
```

$\partial_x f = \frac{2 x y^3}{1 + x^4 y^6}$
$\partial_y f = \frac{3 x^2 y^2}{1 + x^4 y^6}$

i)

```
f[x_, y_] = x + x y^2 + Log[Sin[x^2 + y]];
```

$\partial_x f = 1 + y^2 + 2 x \text{ Cot}[x^2 + y]$
$\partial_y f = 2 x y + \text{Cot}[x^2 + y]$

j)

```
f[x_, y_, z_] = z Exp[x^2 + y^2];
```

$\partial_x f = 2 e^{x^2+y^2}\ x\ z$
$\partial_y f = 2 e^{x^2+y^2}\ y\ z$
$\partial_z f = e^{x^2+y^2}$

k)

```
f[x_, y_, z_] = Log[Exp[x] + z^y];
```

$\partial_x f = \frac{e^x}{e^x + z^y}$
$\partial_y f = \frac{z^y \,\mathrm{Log}[z]}{e^x + z^y}$
$\partial_z f = \frac{y\, z^{-1+y}}{e^x + z^y}$

l)

```
f[x_, y_, z_] = (x y^3 + Exp[z]) / (x^3 y - Exp[z]);
```

$\partial_x f = -\frac{y\left(2 x^3 y^3 + e^z \left(3 x^2 + y^2\right)\right)}{\left(e^z - x^3 y\right)^2}$
$\partial_y f = \frac{2 x^4 y^3 - e^z x \left(x^2 + 3 y^2\right)}{\left(e^z - x^3 y\right)^2}$
$\partial_z f = \frac{e^z x y \left(x^2 + y^2\right)}{\left(e^z - x^3 y\right)^2}$

Exercício 3.3

a)

$\partial_x f(0,0) =$ Não existe
$\partial_y f(0,0) =$ Não existe

b)

$\partial_x f(0,0) = 0$
$\partial_y f(0,0) = 0$

Exercício 3.4

a)

```
f[x_, y_] = Exp[x y];
```

$x\partial_x f = e^{x y}\ x\ y$
$y\partial_y f = e^{x y}\ x\ y$

b)

```
f[x_, y_] = Log[x^2 + y^2 + x y];
```

$x\partial_x f = \frac{x\,(2 x + y)}{x^2 + x y + y^2}$
$y\partial_y f = \frac{y\,(x + 2 y)}{x^2 + x y + y^2}$

$x\partial_x f + y\partial_y f = 2$

c)

```
f[x_, y_, z_] = x + (x - y) / (y - z);
```

$\partial_x f = 1 + \frac{1}{y - z}$
$\partial_y f = \frac{-x + z}{(y - z)^2}$
$\partial_z f = \frac{x - y}{(y - z)^2}$

$\partial_x f + \partial_y f + \partial_z f = 1$

Exercício 3.5

```
Ddirecao[f_, P_, u_] := Limit[(f @@ (P + h u) - f @@ P) / h, h → 0];
```

a)

```
f[x_, y_] = x^2 y + x; P = {1, 0}; u = {1, 1};

Ddirecao[f, P, Normalize[u]]
```

$\sqrt{2}$

b)

```
f[x_, y_] = x^2 Sin[2 y]; P = {1, π/2}; u = {3, -4};

Ddirecao[f, P, Normalize[u]]
```

$\frac{8}{5}$

c)

```
f[x_, y_, z_] = x^2 + y^2 + z^2; P = {1, 2, 3}; u = {1, 1, 1};

Ddirecao[f, P, Normalize[u]]
```

$4\sqrt{3}$

Exercício 3.6

a)

```
Grad[x Exp[-x + y], {x, y}]
```

$\{e^{-x+y} - e^{-x+y}\,x,\ e^{-x+y}\,x\}$

b)

```
Grad[r Sin[θ], {r, θ}]
```

{Sin[θ], r Cos[θ]}

c)

```
Grad[π r^2 h, {r, h}]
```

$\{2\,h\,\pi\,r,\ \pi\,r^2\}$

d)

```
Grad[x Exp[-x^2 - y^2 - z^2], {x, y, z}]
```

$\{e^{-x^2-y^2-z^2} - 2\,e^{-x^2-y^2-z^2}\,x^2,\ -2\,e^{-x^2-y^2-z^2}\,x\,y,\ -2\,e^{-x^2-y^2-z^2}\,x\,z\}$

e)

$\text{Grad}\left[\frac{x\,y\,z}{x^2+y^2+z^2+1}, \{x, y, z\}\right]$ // Simplify

$\left\{\frac{y\,z\,(1-x^2+y^2+z^2)}{(1+x^2+y^2+z^2)^2},\ \frac{x\,z\,(1+x^2-y^2+z^2)}{(1+x^2+y^2+z^2)^2},\ \frac{x\,y\,(1+x^2+y^2-z^2)}{(1+x^2+y^2+z^2)^2}\right\}$

f)

```
Grad[z^2 Exp[x] Cos[y], {x, y, z}] // Simplify
```

$\{e^x\,z^2\,\text{Cos}[y],\ -e^x\,z^2\,\text{Sin}[y],\ 2\,e^x\,z\,\text{Cos}[y]\}$

Exercício 3.7

```
f[x_, y_] = x^2 + y^3;
```

Equação do plano tangente: $z = -11 + 6\,x + 3\,y$

Exercício 3.8

```
f[x_, y_] = x^2 + y^2;
```

Equação do plano tangente a f em (0,0): $z = 0$

```
g[x_, y_] = -x^2 - y^2 + x y^3;
```

Equação do plano tangente a g em (0,0): $z = 0$

Exercício 3.9

```
f[x_, y_] = Exp[x^2 - y^2];
```

a)

Equação do plano tangente a f em (1,1): $z = 1 + 2\,x - 2\,y$

Ponto de interseção do plano tangente com o eixo dos zz: $\{\{x \to 0,\ y \to 0,\ z \to 1\}\}$

b)

Cota do ponto do plano tangente correspondente a x=y=1: z=1.24

```
f[1, 1]
```

1

Exercício 3.10

Ver diapositivos

Exercício 3.11

Ver diapositivos

Exercício 3.12

a)

```
f[x_, y_] = x^2 y/(x^2 + y^2);
f[0, 0] = 0;
```

$$Df((0,0);(u1,u2)) = \frac{u1^2\,u2}{u1^2 + u2^2}$$

Se $(x,y) \neq (0,0)$ então $Df((x,y);(u1,u2)) = \frac{x\left(2\,u1\,y^3 + u2\left(x^3 - x\,y^2\right)\right)}{\left(x^2 + y^2\right)^2}$

b)

f não é diferenciável na origem

Exercício 3.13

a)

```
f[x_, y_] = (x^2 + y^2) Sin[1/Sqrt[x^2 + y^2]];
f[0, 0] = 0;
```

$\partial_x f(0,0) = 0$

$\partial_y f(0,0) = 0$

b)

$$\partial_x f(x,y) = -\frac{x\,\mathrm{Cos}\left[\frac{1}{\sqrt{x^2+y^2}}\right]}{\sqrt{x^2 + y^2}} + 2\,x\,\mathrm{Sin}\left[\frac{1}{\sqrt{x^2 + y^2}}\right]$$

$$\partial_y f(x,y) = -\frac{y\,\mathrm{Cos}\left[\frac{1}{\sqrt{x^2+y^2}}\right]}{\sqrt{x^2 + y^2}} + 2\,y\,\mathrm{Sin}\left[\frac{1}{\sqrt{x^2 + y^2}}\right]$$